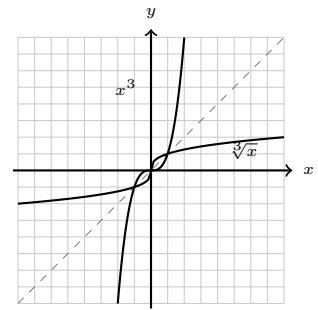


Parent Functions & Key Points**Cubic:** $f(x) = x^3$

x	-2	-1	0	1	2
y	-8	-1	0	1	8

Cube Root: $f(x) = \sqrt[3]{x}$

x	-8	-1	0	1	8
y	-2	-1	0	1	2

Inverses Graph**Remember:** Inverse graphs reflect over the line _____. Points (x, y) become _____.**4-Step Inverse Method**

1. Replace $f(x)$ with y
2. Swap x and y
3. Solve for y (Isolate the power first!)
4. Write using $f^{-1}(x)$ notation

Key Property**The "Un-doing" Relationship:**

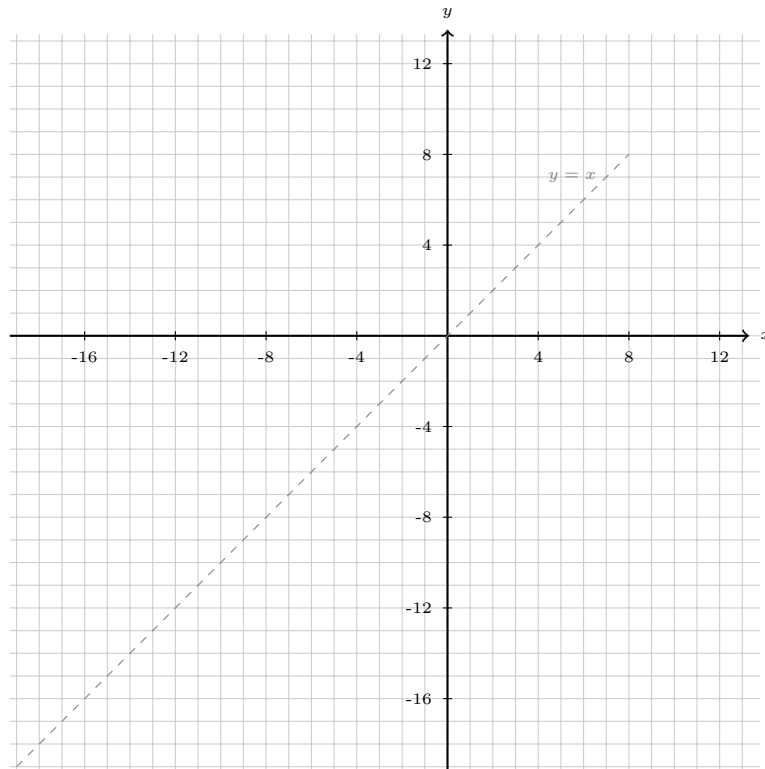
- To undo x^3 , use $\sqrt[3]{x}$
 - To undo $\sqrt[3]{x}$, use x^3
- Example: $\sqrt[3]{x^3} = x$

Example 1: $f(x) = 2x^3 - 3$ *(Vertical stretch by 2, down 3)***Find the inverse:****Graph both:**

$$f^{-1}(x) = \underline{\hspace{2cm}}$$

Key points:

$f(x) = 2x^3 - 3$	$f^{-1}(x)$
$(-2, -19)$	
$(-1, -5)$	
$(0, -3)$	
$(1, -1)$	
$(2, 5)$	



Big Idea: When solving, always **isolate** the term with the exponent or radical *before* you take the root or power!

Algebraic Practice: Find the inverse using the 4-step method.

Example 4: $f(x) = 4(x - 2)^3$

Example 5: $f(x) = -2\sqrt[3]{x + 6}$

$f^{-1}(x) =$ _____

$f^{-1}(x) =$ _____

Example 6: $f(x) = x^3 + 4$

(Shifted Up 4)

Find the inverse:

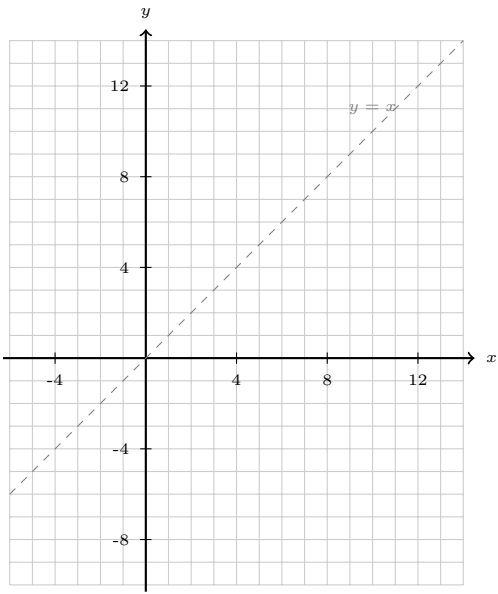
Graph both:

$f^{-1}(x) =$ _____

Key points: Start with parent, shift up 4

Parent	$f(x)$
$(-2, -8)$	$(-2, \quad)$
$(-1, -1)$	$(-1, \quad)$
$(0, 0)$	$(0, \quad)$
$(1, 1)$	$(1, \quad)$
$(2, 8)$	$(2, \quad)$

$f(x)$	$f^{-1}(x)$
$(\quad, \quad)$	$(\quad, \quad)$
$(\quad, \quad)$	$(\quad, \quad)$
$(\quad, \quad)$	$(\quad, \quad)$
$(\quad, \quad)$	$(\quad, \quad)$
$(\quad, \quad)$	$(\quad, \quad)$



Example 7: $f(x) = \sqrt[3]{x + 8}$

(Shifted Left 8)

Find the inverse:

Graph both:

$f^{-1}(x) =$ _____

Key points: Start with parent, shift left 8

Parent	$f(x)$
$(-8, -2)$	$(\quad, -2)$
$(-1, -1)$	$(\quad, -1)$
$(0, 0)$	$(\quad, 0)$
$(1, 1)$	$(\quad, 1)$
$(8, 2)$	$(\quad, 2)$

$f(x)$	$f^{-1}(x)$
$(\quad, \quad)$	$(\quad, \quad)$
$(\quad, \quad)$	$(\quad, \quad)$
$(\quad, \quad)$	$(\quad, \quad)$
$(\quad, \quad)$	$(\quad, \quad)$
$(\quad, \quad)$	$(\quad, \quad)$

